

The cost factor (capital expenditure)

Capital costs of quantum internet are dominated by such things as specialised photonic infrastructure, cryogenics, precision timing, and quantum-grade endpoints, not bandwidth. Low-loss fibre paths, wavelength-stable lasers, ultra-low-noise detectors (SNSPDs are often cryogenically cooled), and high-extinction modulators must be installed at the physical layer, and are expensive, particularly when extending (metro/backbone) networks to accommodate quantum channels in addition to classical traffic. Meanwhile, trusted-node QKD networks are available for deployment today but incur capital expenditure in secure facilities, tamper-resistant hardware, and operational hardening at every intermediate node — costs that increase linearly with distance and node count.

The long-term cost inflection is in repeated entanglement. True quantum repeaters need quantum memories with long coherence time, high retrieval efficiency, and telecom-compatible photonic interfaces. Because these nodes often require vacuum systems, cryostats or ultra-stable environments, single-photon sources or entangled-pair sources, frequency conversion, and high-speed classical control electronics, their construction is not trivial. Networks also require precision time/frequency distribution (GPS-disciplined clocks might be too coarse for some synchronisation targets), as well as calibration and monitoring instrumentation to keep interference visibility and polarisation stability. Capital expenditure is also influenced by the deployment model: metro scale testbeds can collocate in ducts and data-centre footprints, whereas intercity links may need dedicated dark fibre or meticulously engineered coexistence with classical DWDM traffic to reduce Raman noise. Finally, maturity in manufacturing has an impact: the

costs are high since many components (memories, detectors, integrated photonics) are manufactured in small volumes, with yield and packaging—particularly fibre coupling and cryogenic reliability—frequently turning out to be the secret cost drivers.

Future of quantum internet development

It is anticipated that near-term development of the quantum internet will be rolled out in stages, with functionalities being added as hardware evolves. Phase one centres on quantum-secure communications, mainly QKD (including measurement-device-independent variants) and quantum-randomness services, offered through metro networks and limited intercity routes. At the same time, a second line of research addresses entanglement-as-a-service: networks that allocate entangled pairs between end nodes with given fidelity and rate, allowing teleportation-based primitives and fundamental multi-node experiments (entanglement swapping chains, simplest quantum routing, and networked sensing).

A critical transition will be from ‘trusted-node’ scaling to repeater-based scaling, motivated by progress in quantum memories, integrated

photonics, and error mitigation. Architecturally, the field is moving to a layered network stack, including explicit link-layer heralding, purification/encoding choices, and a control plane that schedules in the face of constraints (memory lifetimes, probabilistic link establishment, classical feed-forward latency). Multiplexing—temporal, spectral, and spatial—will be important to increase the rates of entanglement generation, and hardware-protocol co-design will define what are practical throughputs.

In the long run, the quantum internet is expected to link quantum processors for distributed computing and secure delegated computation, and enable quantum-enhanced sensing (clock networks, interferometric baselines, and correlated measurements). Success will require interfaces and metrics to be standardised, telecom-wavelength compatible node technologies to mature, and real-world operation to be demonstrated. The ‘end state’ is not one giant monolithic network but rather a federation of quantum-capable domains—metro, backbone, and satellite—that are integrated with classical infrastructure via well-articulated service-level objectives for entanglement and security. **END** 

References

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